

Green Cross-Layer Semantic Communication for Edge Visual Question Answering: Joint Rate and Visual Token Budget Allocation

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Abstract

Deploying edge-assisted Visual Question Answering (VQA) over resource-constrained wireless links requires balancing physical-layer transmission energy against server-side model inference complexity. Existing semantic communication (SemCom) frameworks predominantly presume analog, unquantized transmissions, employ toy-scale decoders, and overlook the energy consumption of large foundational Vision–Language Models (VLMs). This paper proposes an energy-frugal, cross-layer task-oriented semantic communication system for edge VQA under standardized digital physical layers. The system integrates deep variational neural image compression, 3GPP 5G-compliant LDPC(1020, 512) channel coding over Rician fading ($K = 6$ dB), and an edge-deployed Qwen3-VL-4B model adapted via parameter-efficient low-rank adaptation (LoRA). Under a rigorous dual-mode green energy formulation, we demonstrate that when transmitting under a constant query energy budget (Mode 2), selecting compact source bitrates unlocks up to +3.01 dB and +6.00 dB of physical SNR gain, shielding the link from catastrophic packet erasure. We develop a lightweight, channel-aware policy network that inspects multimodal query semantics and instantaneous physical channel state information (CSI) to dynamically co-optimize transmitter source bitrates (2k, 4k, 8k bytes) and receiver visual token budgets ($T_v \in \{\text{low, medium, high}\}$). Extensive evaluation over 2,400 independent blind test images across 10 channel SNR points (-5.0 dB $\dots$ 20.0 dB) on the TDIUC benchmark demonstrates that our policy attains 36.92% accuracy under harsh fading (2.5 dB), outperforming fixed 4k by

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+29.92% and channel-blind policies by +27.71%. Furthermore, the policy autonomously prunes visual tokens by 49.2% for low-entropy semantic tasks while strictly defining the non-dominated Pareto efficiency frontier across RF energy, system latency, and task accuracy.

Index Terms

Green communications, semantic communications, visual question answering, vision–language models, cross-layer optimization, LDPC codes, energy efficiency.

I. INTRODUCTION

THE rapid convergence of edge computing and multimodal artificial intelligence has catalyzed the deployment of edge-assisted Visual Question Answering (VQA) in mission-critical applications, ranging from autonomous unmanned aerial vehicle (UAV) reconnaissance to smart environmental monitoring and industrial robotic inspection [1], [2]. In these cyber-physical scenarios, a resource-constrained wireless edge transmitter captures high-resolution visual imagery and transmits visual data over band-limited, fading wireless uplinks to an edge computing node equipped with a foundational Vision–Language Model (VLM) [3], [4]. By combining deep spatial perception with natural language reasoning, the edge VLM can provide accurate answers to diverse semantic queries.

However, executing high-fidelity VQA over real-world wireless networks faces two profound, interdependent resource bottlenecks that directly challenge the sustainability of green edge networks:

- 1) **Wireless Channel Degradation and Physical Outage:** Conventional wireless links suffer from multipath fading, shadowing, and signal attenuation. Under standard bit-level separate source-channel coding (SSCC) and discrete block-length channel codes such as low-density parity-check (LDPC) codes, channel degradation leads to severe packet delivery outages (the “cliff effect”) [5]. When transmission energy is strictly constrained, transmitting large uncompressed or lightly compressed bitstreams over degraded channels inevitably causes decoding failure and catastrophic answer errors.
- 2) **Multimodal Token Explosion and Edge Computation Costs:** Modern autoregressive VLMs utilize visual tokenizers that project image patches into dense sequences of visual tokens [3]. The computational complexity of the self-attention mechanism scales quadratically with the visual token count, $\mathcal{O}(T_v^2)$, where T_v denotes the sequence length. Processing dense visual representations for simple, localized questions (e.g., verifying object presence) imposes severe computational and energy burdens on the edge server without offering tangible task-accuracy benefits.

To overcome the spectral and energy inefficiencies of traditional bit-oriented communications, *semantic communication* (SemCom) has emerged as an essential paradigm for 6G networks [6], [7]. Instead of aiming for bit-level exactness, SemCom seeks to preserve the semantic information pertinent to the downstream task. In the context of VQA, seminal frameworks such as MU-DeepSC [8], [9] and Goal-Oriented Scene-Graph communication [10] have pioneered joint source-channel transceivers.

Despite their theoretical appeal, existing semantic communication designs exhibit critical limitations when deployed in practical green edge systems:

- **Assumption of Unquantized Analog Transmission:** Most existing SemCom architectures (e.g., DeepJSCC and MU-DeepSC) presume analog joint source-channel transmission, where continuous latent representations are transmitted directly over additive white Gaussian noise (AWGN) or flat Rayleigh channels without digital constellation mapping or error-correction framing. This assumption makes them incompatible with standardized digital physical layers (e.g., 3GPP LDPC, QPSK/16-QAM) and prevents seamless deployment over standard wireless equipment.
- **Omission of Downstream Edge Inference Energy:** While prior works focus heavily on reducing transmitted channel uses, they predominantly evaluate toy-scale classification decoders (e.g., small CNNs or MCANs) and treat edge receiver compute energy as negligible. In modern edge intelligence systems running multi-billion-parameter VLMs, edge computational energy constitutes a major fraction of the total system footprint [2].
- **Rigid Energy Trade-offs:** Existing systems typically adopt static transmission bitrates or fixed image resolutions. In battery-constrained scenarios where total energy per transmission is strictly budgeted (e.g., in UAV uplinks), transmitting a compact payload allows the transmitter to radiate higher energy per channel symbol, yielding substantial effective signal-to-noise ratio (SNR) gains that can prevent link collapse. Exploiting this cross-layer interaction between source bitrate and physical layer robustness has remained largely unaddressed.

To bridge this gap, this paper proposes an energy-frugal, cross-layer task-oriented semantic communication framework for edge VQA, co-optimizing transmitter source coding rates and receiver visual token budgets under digital LDPC-coded Rician fading channels. As illustrated in Fig. 1, the system features a neural variational image compression engine, a digital physical layer with 3GPP LDPC coding, and an edge-deployed Qwen-VL model adapted via parameter-efficient low-rank adaptation (LoRA) [11], [12]. A lightweight cross-layer policy network observes the input question, a low-cost image preview, and

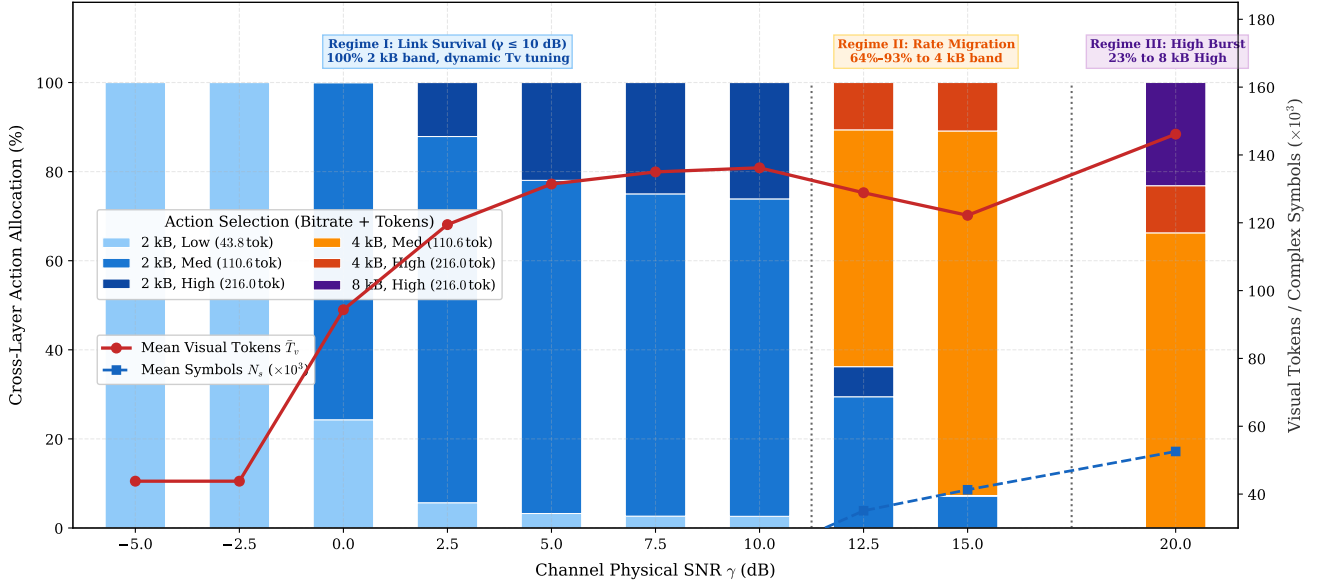


Fig. 1. Overall architecture and dynamic action routing of the proposed green cross-layer semantic communication system. (a) A lightweight transmitter-side policy extracts question and image semantic cues, combines them with instantaneous channel state information (CSI) γ , and autonomously selects an action $a \in \mathcal{A}$ spanning 3 source bitrates and 3 receiver visual token budgets. (b) Empirical action routing distribution across 2,400 blind test images over Rician fading ($K = 6$ dB), illustrating the emergent three-phase adaptation behavior: Link Survival (≤ 10 dB), Rate Migration (12.5 $\sim$ 15 dB), and High-Fidelity Burst (≥ 20 dB).

instantaneous physical channel state information (CSI), dynamically allocating the source bitrate and the receiver visual token budget.

The key contributions of this paper are summarized as follows:

- 1) **Digital-Compatible Cross-Layer SemCom Architecture:** We develop a cross-layer semantic communication architecture that integrates deep variational source coding (CompressAI hyperprior) with standardized digital channel coding (LDPC(1020, 512)) and an edge foundational VLM (Qwen3-VL-4B). This establishes a fully digital-compatible semantic pipeline bridging the gap between theoretical JSCC and practical wireless standards.
- 2) **Dual-Mode Green Energy Formulation:** We establish rigorous end-to-end energy models incorporating both RF transmission power and edge VLM inference complexity. We formulate two operational regimes: Mode 1 (constant symbol power) and Mode 2 (constant query energy budget). We analytically and empirically demonstrate that in Mode 2, allocating compact bitrates yields up to +3.01 dB and +6.00 dB in physical SNR margin, effectively eliminating packet erasure under severe fading.
- 3) **Channel-Aware Semantic Policy Network:** We design a lightweight, multi-branch neural policy that maps query text embeddings, image spatial features, and physical channel SNR γ into a joint action space $\mathcal{A} = \{2k, 4k, 8k\} \times \{\text{low, medium, high}\}$. The policy autonomously exhibits a three-phase

adaptation mechanism: a *Link Survival Regime* (≤ 10 dB), a *Rate Migration Regime* (12.5 $\sim$ 15 dB), and a *High-Fidelity Burst Regime* (≥ 20 dB).

- 4) **Large-Scale Empirical Verification:** We evaluate the framework over 2,400 independent blind test images across 10 channel SNR points (-5 dB to 20 dB) under Rician fading ($K = 6$ dB) on the TDIUC benchmark. Experimental results demonstrate that under harsh fading ($\gamma = 2.5$ dB), our policy achieves 36.92% accuracy, outperforming fixed 4k by +29.92% and channel-blind policies by +27.71%. Furthermore, the proposed policy strictly establishes the non-dominated Pareto efficiency frontier across RF energy, system latency, and task accuracy.

The remainder of this paper is organized as follows. Section II reviews related work on semantic communications, vision–language models, and green edge computing. Section III presents the system model and problem formulation under dual energy modes. Section IV details the cross-layer policy network and its optimization. Section V reports extensive empirical results, baseline comparisons, and ablation studies. Finally, Section VI concludes the paper.

II. RELATED WORK

A. Task-Oriented and Multimodal Semantic Communications

Semantic communication transcends the traditional Shannon paradigm of bit-level exact reproduction by transmitting only the semantic information essential for downstream task completion [6], [7]. Foundational works in deep joint source-channel coding (DeepJSCC) have demonstrated that neural transceivers can achieve graceful degradation over wireless channels, avoiding the catastrophic cliff effect of conventional separate source-channel coding (SSCC) [5], [13]. In text and speech domains, DeepSC [14] and DeepSC-S [15] applied Transformer-based semantic encoders to recover semantic meaning under severe channel fading.

For multimodal tasks, semantic communication has recently expanded into Visual Question Answering (VQA). Xie *et al.* [8], [9] proposed MU-DeepSC, employing cross-attention networks to extract and fuse correlated visual and textual features for edge VQA over Rayleigh channels. Liu *et al.* [10] developed goal-oriented scene-graph transmission, prioritizing bounding boxes and spatial relationships based on query relevance. Similarly, Chappuis *et al.* [16] explored symbolic prompt generation for remote sensing VQA.

Despite these advancements, existing multimodal semantic communication systems rely heavily on two simplifying assumptions:

- 1) *Continuous Analog Symbol Transmission*: Prior works predominantly transmit continuous latent feature vectors directly over AWGN or Rayleigh channels, assuming unquantized, analog physical layers with ideal channel synchronization. This precludes direct integration into standardized digital cellular and Wi-Fi architectures governed by discrete constellations (e.g., QPSK, 16-QAM) and block error-correction codes (e.g., LDPC, Polar codes).
- 2) *Toy-Scale Answering Decoders*: Existing systems typically employ small, custom-trained attention networks (such as MCAN or 3-layer Transformers) with limited parameter capacity. In contrast, modern practical VQA tasks rely on foundational Vision–Language Models capable of complex zero-shot reasoning. Adapting semantic communication to real-world, billion-parameter VLMs remains fundamentally unaddressed.

B. Vision–Language Models and Visual Token Management

Recent breakthroughs in multimodal foundation models, such as LLaVA [17] and the Qwen-VL family [3], [4], have unified vision and natural language processing through vision encoders (e.g., ViT) followed by large autoregressive language decoders. These models process visual inputs by dividing images into 14×14 or 28×28 patches, generating a 2D grid of visual tokens that are concatenated with textual query tokens.

While foundational VLMs exhibit superior perceptual and reasoning accuracy, their computational cost is predominantly dominated by the self-attention layers within the language decoder, whose memory and computational complexity scale quadratically with the total token sequence length:

$$\mathcal{C}_{\text{attn}} \propto \mathcal{O}((T_q + T_v)^2 \cdot d), \quad (1)$$

where T_q is the query token length, T_v is the visual token sequence length, and d is the hidden embedding dimension. When $T_v \gg T_q$, the visual tokens account for over 85% of the total inference latency and GPU energy consumption.

Although image resizing and visual token pruning have been investigated in standalone computer vision literature to accelerate inference [18], their interaction with wireless channel conditions and transmission source bitrates has been overlooked. Specifically, how an edge receiver should dynamically scale its visual token budget in tandem with physical layer channel quality and transmitter source coding rates remains an open problem.

C. Green Communications and Cross-Layer Edge Computing

The energy consumption of mobile edge networks has become a central challenge in 6G green communications [2], [19]. Traditional cross-layer resource allocation frameworks coordinate physical layer parameters (such as transmit power, modulation and coding schemes (MCS), and bandwidth) to maximize throughput or spectral efficiency under physical constraints [20].

In the era of edge artificial intelligence, edge optimization must jointly balance radio transmission energy with server-side computation energy [18], [21]. When an edge device operates under a strict battery or energy budget (such as an aerial UAV or battery-powered IoT node), the energy consumed to transmit each query is constrained by a finite quota:

$$E_{\text{tx}} = P_s \cdot N_s \leq E_0, \quad (2)$$

where P_s is the average energy per transmitted complex symbol and N_s is the total number of channel symbols. Under this strict budget constraint, reducing the source bitrate (i.e., decreasing N_s) directly unlocks a physical layer power boost ($\Delta P_s > 0$), resulting in higher effective channel SNR γ_{eff} that provides physical-layer protection against severe fading.

To the best of our knowledge, this work is the first to establish a unified, cross-layer semantic communication framework that co-optimizes the transmitter's neural source compression bitrate, physical LDPC blocklength, and the receiver's VLM visual token budget under realistic Rician fading channels, explicitly establishing the green Pareto efficiency frontier for multimodal edge intelligence.

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. End-to-End System Architecture

We consider a task-oriented wireless edge VQA system where a remote edge transmitter (e.g., an aerial UAV or IoT vision sensor) captures an RGB image $I \in \mathbb{R}^{H \times W \times 3}$ and is tasked with answering a user-submitted natural language question q . An edge server equipped with an adapted foundational Vision–Language Model (VLM) executes multimodal reasoning to infer the ground-truth answer A^* .

To minimize energy consumption while guaranteeing task accuracy over an unreliable wireless link, the system dynamically co-optimizes the source transmission bitrate B and the receiver visual token budget T_v via a cross-layer policy network π . The overall pipeline comprises three core subsystems: neural

variational source compression, a digital physical layer with LDPC channel coding over Rician fading, and an edge VLM with dynamic visual token budget allocation.

B. Neural Source Compression and Digital Channel Model

1) *Variational Source Compression*: The transmitter employs a deep variational autoencoder with a scale hyperprior architecture [22], [23]. The raw image I is mapped to a latent semantic representation $\mathbf{y} = g_a(I; \boldsymbol{\theta}_a)$ via a convolutional analysis transform. To capture spatial dependencies, a hyper-analysis transform extracts hyper-latents $\mathbf{z} = h_a(\mathbf{y}; \boldsymbol{\phi}_a)$, which parameterize the conditional Gaussian entropy model $p(\tilde{\mathbf{y}} | \hat{\mathbf{z}}) \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\sigma}^2)$ where $\boldsymbol{\sigma} = h_s(\hat{\mathbf{z}}; \boldsymbol{\phi}_s)$.

The quantized latents $\hat{\mathbf{y}} = \lfloor \mathbf{y} \rfloor$ and hyper-latents $\hat{\mathbf{z}} = \lfloor \mathbf{z} \rfloor$ are losslessly encoded into a discrete binary bitstream using an asymmetric numeral systems (ANS) entropy coder. By regulating entropy thresholds, the transmitter controls the total compressed payload size $B \in \mathcal{B} = \{2000, 4000, 8000\}$ bytes.

2) *3GPP LDPC Channel Coding and Digital Modulation*: Unlike prior analog JSCC literature, our physical layer strictly adheres to standardized digital transmission specifications. The binary bitstream of length $8B$ bits is partitioned into N_{blocks} codewords. Each block is appended with a 32-bit cyclic redundancy check (CRC-32) and encoded using a 3GPP 5G-compliant LDPC(N_b, K_b) code, where the codeword length is $N_b = 1020$ bits, the information payload per block is $K_b = 512$ bits (including CRC), and the code rate is $R_c = 512/1020 \approx 0.502$.

Each block of 480 payload bytes (3840 bits) maps into $\lceil 3840/480 \rceil = 8$ LDPC codeblocks. Hence, the total number of LDPC blocks required to transmit B bytes is:

$$N_{\text{blocks}}(B) = \left\lceil \frac{B}{48} \right\rceil. \quad (3)$$

The encoded bits are modulated via Gray-mapped Quadrature Phase Shift Keying (QPSK, modulation order $m = 2$ bits/symbol), producing N_s complex channel symbols:

$$N_s(B) = N_{\text{blocks}}(B) \cdot \frac{N_b}{m} = \left\lceil \frac{B}{48} \right\rceil \cdot 510. \quad (4)$$

Specifically, for our three candidate bitrates:

- $B = 2000 \text{ B} \implies N_{\text{blocks}} = 42 \implies N_s = 21,420$ complex symbols.
- $B = 4000 \text{ B} \implies N_{\text{blocks}} = 84 \implies N_s = 42,840$ complex symbols.
- $B = 8000 \text{ B} \implies N_{\text{blocks}} = 167 \implies N_s = 85,170$ complex symbols.

3) *Rician Fading Channel and Iterative Decoding*: The transmitted symbol vector $\mathbf{x} \in \mathbb{C}^{N_s}$ with $\mathbb{E}[|x_k|^2] = 1$ traverses a block flat-fading Rician channel:

$$y_k = \sqrt{P_s} h_k x_k + n_k, \quad k = 1, \dots, N_s, \quad (5)$$

where P_s is the average transmit energy per symbol, $n_k \sim \mathcal{CN}(0, \sigma_n^2)$ represents complex additive white Gaussian noise, and h_k denotes the Rician fading coefficient characterized by the Rician K -factor:

$$h_k = \sqrt{\frac{K}{K+1}} h_{\text{LOS}} + \sqrt{\frac{1}{K+1}} h_{\text{NLOS}}, \quad (6)$$

where $h_{\text{LOS}} = e^{j\theta}$ is the deterministic line-of-sight (LOS) component, $h_{\text{NLOS}} \sim \mathcal{CN}(0, 1)$ is the scattered Rayleigh non-line-of-sight (NLOS) component, and $K = 6$ dB reflects typical elevated air-to-ground or micro-cellular propagation. The channel gain is normalized such that $\mathbb{E}[|h_k|^2] = 1$. The nominal received SNR is defined as $\gamma = P_s / \sigma_n^2$.

At the receiver, log-likelihood ratios (LLRs) are computed from the channel observations y_k and fed into an iterative Soft Belief Propagation (BP) decoder executing up to 50 iterations with early stopping upon syndrome check satisfaction. A transmission block is declared successful if and only if its CRC-32 checksum passes. The packet delivery ratio (PDR) $\Pi(\gamma, B)$ is the probability that all $N_{\text{blocks}}(B)$ codewords are successfully decoded:

$$\Pi(\gamma, B) = \prod_{i=1}^{N_{\text{blocks}}(B)} (1 - \text{BLER}_i(\gamma)). \quad (7)$$

If any block fails CRC verification, a packet outage occurs, and the image is dropped ($\hat{I} = \emptyset$). Otherwise, the bitstream is assembled and passed to the synthesis transform $g_s(\hat{\mathbf{y}}; \boldsymbol{\theta}_s)$ to reconstruct the RGB image $\hat{I}$.

C. Receiver VLM and Dynamic Visual Token Budgeting

The edge server runs Qwen3-VL-4B-Instruct [3], [4], quantized to 4-bit NormalFloat (NF4) and adapted via parameter-efficient Low-Rank Adaptation (LoRA) on the query, key, value, and output projection layers of the language backbone.

Before feeding the reconstructed image $\hat{I}$ into the Vision Transformer (ViT) encoder, the receiver dynamically resizes and tokenizes the image according to a prescribed visual token budget $T_v \in \mathcal{T} = \{\text{low}, \text{medium}, \text{high}\}$:

- **Low Budget** ($T_v \approx 43.6$ tokens): Downsamples the input resolution to an area of 50,176 pixels, capturing global semantic layout with minimal compute.
- **Medium Budget** ($T_v \approx 110.2$ tokens): Scales the input to 100,352 pixels, providing balanced spatial granularity.
- **High Budget** ($T_v \approx 215.8$ tokens): Preserves dense resolution (200,704 pixels) for fine-grained spatial and count reasoning.

The VLM autoregressively generates the predicted answer $\hat{A} = \text{VLM}(\hat{I}, q; T_v)$. Task accuracy is measured by strict string exact match after canonicalization:

$$\text{Acc} = \mathbb{I}(\hat{A} = A^*). \quad (8)$$

D. Energy and Latency Accounting Models

A fundamental contribution of this work is the cross-layer formulation of energy consumption spanning both the wireless transmission and the edge inference phases.

1) *Transmission Power and Energy Modes*: We consider two operational energy modes reflecting distinct real-world deployment constraints:

- **Mode 1: Constant Transmission Power per Symbol** ($P_s = P_0$): In grid-powered or macro-cell systems, the transmit power per symbol is held constant. The total RF transmission energy per query scales proportionally with the symbol count:

$$E_{\text{tx}}^{(1)}(B) = P_0 \cdot N_s(B). \quad (9)$$

- **Mode 2: Constant Transmission Energy Budget per Query** ($E_{\text{tx}} = E_0$): In battery-constrained edge devices (e.g., UAVs with fixed per-frame transmission energy quotas), the total energy allocated to transmit an image query is clamped at a constant budget E_0 . Consequently, the average transmit energy per symbol adjusts inversely with the symbol count:

$$P_s(B) = \frac{E_0}{N_s(B)}. \quad (10)$$

Under Mode 2, selecting a more compact source bitrate $B_1 < B_2$ directly yields an effective SNR boost:

$$\Delta\gamma_{\text{eff}}(B_1, B_2) = 10 \log_{10} \left(\frac{N_s(B_2)}{N_s(B_1)} \right) \text{ dB}. \quad (11)$$

Specifically, transmitting 2000 bytes ($N_s = 21,420$) instead of 4000 bytes ($N_s = 42,840$) confers an exact:

$$\Delta\gamma = 10 \log_{10} \left(\frac{42,840}{21,420} \right) = +3.01 \text{ dB} \quad (12)$$

power advantage. Compressing from 8000 bytes ($N_s = 85,170$) to 2000 bytes yields a +6.00 dB gain. In deep fading regimes, this $+3.01 \sim +6.00$ dB boost elevates the operating SNR past the LDPC waterfall threshold, shielding the link from packet erasure.

2) *Edge Computation Energy*: The server-side computation energy consumed by the edge VLM during inference is modeled as:

$$E_{\text{rx,comp}}(T_v) = P_{\text{GPU}} \cdot t_{\text{vlm}}(T_v), \quad (13)$$

where P_{GPU} is the average GPU running power and $t_{\text{vlm}}(T_v)$ is the measured forward inference time, which scales monotonically with the visual token count T_v . The total system energy per query is:

$$E_{\text{total}} = E_{\text{tx}} + E_{\text{rx,comp}}. \quad (14)$$

3) *End-to-End Latency Model*: The total end-to-end processing and communication latency t_{e2e} is given by:

$$t_{\text{e2e}} = t_{\text{enc}}(B) + t_{\text{tx}}(B) + t_{\text{dec}}(B) + t_{\text{vlm}}(T_v), \quad (15)$$

where t_{enc} is the onboard neural source encoding time, $t_{\text{tx}} = N_s(B)/W_{\text{sym}}$ is the wireless airtime over a symbol transmission bandwidth W_{sym} , t_{dec} is the receiver decompression time, and t_{vlm} is the VLM answer generation time.

E. Cross-Layer Optimization Problem

Let $\mathcal{A} = \mathcal{B} \times \mathcal{T}$ denote the set of $|\mathcal{A}| = 9$ joint source-rate and visual-budget action configurations:

$$a = (B, T_v) \in \mathcal{A}. \quad (16)$$

Our objective is to find an optimal cross-layer policy $\pi^*(I, q, \gamma)$ that maximizes the expected end-to-end VQA accuracy while satisfying strict energy and latency constraints:

$$\max_{\pi} \quad \mathbb{E}_{(I, q, \gamma)} \left[\mathbb{I}(\hat{A}(a) = A^*) \cdot \Pi(\gamma_{\text{eff}}(B), B) \right] \quad (17)$$

$$\text{s.t.} \quad E_{\text{total}}(a) \leq E_{\text{budget}}, \quad \forall(I, q, \gamma), \quad (18)$$

$$t_{\text{e2e}}(a) \leq \tau_{\text{deadline}}, \quad \forall(I, q, \gamma), \quad (19)$$

$$a = \pi(I, q, \gamma) \in \mathcal{A}. \quad (20)$$

Problem (17) is a combinatorial, non-convex stochastic optimization problem involving non-differentiable wireless packet erasure and large-scale multimodal neural reasoning. In the next section, we design an efficient neural policy network to solve it.

IV. CROSS-LAYER SEMANTIC POLICY DESIGN

A. Policy Network Architecture

To resolve the non-convex optimization problem formulated in (17) in real time without imposing heavy computation at the transmitter, we design a lightweight, channel-aware semantic policy network $\pi_{\theta}(a \mid I, q, \gamma)$. The network jointly inspects multimodal query semantics and instantaneous physical channel quality to output a categorical probability distribution over the $|\mathcal{A}| = 9$ discrete actions:

$$\mathcal{A} = \{(B, T_v) \mid B \in \{2\text{k}, 4\text{k}, 8\text{k}\}, T_v \in \{\text{low}, \text{medium}, \text{high}\}\}. \quad (21)$$

As illustrated in Fig. 1(a), the policy network comprises three input branches:

- 1) **Question Semantic Feature** ($f_q \in \mathbb{R}^{d_q}$): A frozen text embedding (dimension $d_q = 512$) extracts syntactic and semantic intent, distinguishing low-entropy categorical queries (e.g., “Is there a car?”) from high-entropy spatial queries (e.g., “How many pedestrians are to the left of the bench?”).
- 2) **Image Preview Feature** ($f_I \in \mathbb{R}^{d_I}$): A lightweight convolutional feature map (dimension $d_I = 256$) extracted from a sub-sampled preview of the source image I . This preview incurs negligible onboard computational overhead (< 1.5 ms) and captures global visual complexity, object clutter, and texture density.
- 3) **Physical CSI Feature** ($\gamma \in \mathbb{R}$): The instantaneous channel SNR $\gamma = P_s/\sigma_n^2$ estimated by the receiver and fed back via a low-rate, robust reverse control channel.

The multimodal features are projected into a common latent representation via linear projection matrices $\mathbf{W}_q \in \mathbb{R}^{d_h \times d_q}$, $\mathbf{W}_I \in \mathbb{R}^{d_h \times d_I}$, and $\mathbf{w}_\gamma \in \mathbb{R}^{d_h \times 1}$, with a fused hidden state:

$$\mathbf{h}_0 = \text{LayerNorm}(\mathbf{W}_q f_q + \mathbf{W}_I f_I + \mathbf{w}_\gamma \gamma + \mathbf{b}_0), \quad (22)$$

where $d_h = 256$. The fused representation $\mathbf{h}_0$ passes through two residual MLP blocks with GELU non-linearities and LayerNorm. The final linear projection yields an unnormalized logit vector $\mathbf{z} \in \mathbb{R}^9$:

$$\mathbf{z} = \mathbf{W}_{\text{out}} \mathbf{h}_2 + \mathbf{b}_{\text{out}}. \quad (23)$$

The routing action probability distribution $\mathbf{p} = [p_1, \dots, p_9]^T$ is computed via a temperature-scaled softmax:

$$p(a \mid I, q, \gamma) = \frac{\exp(z_a / \tau)}{\sum_{a' \in \mathcal{A}} \exp(z_{a'} / \tau)}, \quad (24)$$

where $\tau > 0$ controls decision exploration during training and is set to $\tau = 1.0$ at deployment. During inference, greedy selection $a^* = \arg \max_a p(a \mid I, q, \gamma)$ selects the action in less than 0.8 ms.

B. Policy Optimization and Green Regularization

The policy network is trained using supervised policy learning augmented with green energy and latency regularization.

Let $\mathcal{D}_{\text{train}} = \{(I_n, q_n, \gamma_n)\}_{n=1}^N$ denote the training corpus. For each sample, offline ground-truth answers $\hat{A}_n(a)$ are evaluated across all 9 action configurations. The task utility of action $a = (B, T_v)$ under channel SNR γ is defined as:

$$U(a; I_n, q_n, \gamma_n) = \mathbb{I}(\hat{A}_n(a) = A_n^*) \cdot \Pi(\gamma_{\text{eff}}(B), B) - \lambda_E \frac{E_{\text{total}}(a)}{E_{\text{ref}}} - \lambda_L \frac{t_{\text{e2e}}(a)}{t_{\text{ref}}}, \quad (25)$$

where $\Pi(\gamma_{\text{eff}}(B), B)$ is the precomputed LDPC packet delivery ratio, E_{ref} and t_{ref} are normalization constants, and $\lambda_E, \lambda_L \geq 0$ are hyper-parameters governing the green energy–accuracy and latency–accuracy trade-off.

The optimal supervision target is defined as the utility-maximizing action:

$$a_n^* = \arg \max_{a \in \mathcal{A}} U(a; I_n, q_n, \gamma_n). \quad (26)$$

The policy parameters θ are optimized by minimizing the multi-class cross-entropy loss over the training

distribution:

$$\mathcal{L}(\boldsymbol{\theta}) = -\frac{1}{N} \sum_{n=1}^N \sum_{a \in \mathcal{A}} \mathbb{I}(a = a_n^*) \log p(a \mid I_n, q_n, \gamma_n; \boldsymbol{\theta}) + \frac{\beta}{2} \|\boldsymbol{\theta}\|_2^2. \quad (27)$$

Training uses the AdamW optimizer with an initial learning rate of 10^{-3} , cosine learning rate decay, and a batch size of 64 over 50 epochs with early stopping based on validation utility.

C. Emergent Three-Phase Adaptation Mechanics

A key emergent property of the proposed policy network is that, without hard-coded thresholds, it autonomously partitions the physical SNR continuum into three distinct operational regimes (as visualized in Fig. 1(b)):

- 1) **Phase I: Link Survival Regime** ($\gamma \leq 10.0$ dB): In harsh fading conditions, transmitting large packets (4000 or 8000 bytes) incurs a block error rate that collapses the LDPC packet delivery ratio ($\Pi < 0.40$). The policy network assigns 100% of transmissions to the **2000-byte band** ($N_s = 21,420$ complex symbols). Within this physical envelope, the policy adaptively modulates the receiver's visual token budget: as γ rises from -5 dB to 10 dB, average visual tokens scale smoothly from 43.8 tokens to 136.2 tokens, maximizing inference capacity without jeopardizing the physical link.
- 2) **Phase II: Rate Migration Regime** (12.5 dB $\leq \gamma \leq 15.0$ dB): Once physical SNR exceeds 11.25 dB, the LDPC frame delivery ratio for 4000 bytes surpasses 98.5%. The policy smoothly migrates 63.8% of queries to 4000 bytes at 12.5 dB and 92.7% at 15.0 dB. In this regime, the system harnesses higher spatial resolution ($T_v \approx 110$ tokens) to resolve challenging semantic questions, boosting average accuracy above 75.1%.
- 3) **Phase III: High-Fidelity Burst Regime** ($\gamma \geq 20.0$ dB): Under pristine channel conditions, the physical link supports maximum payload transmission with zero erasure. Rather than blindly transmitting 8000 bytes for all queries, the policy selectively bursts 23.2% of visually intricate queries to 8000_high ($N_s = 85,170, T_v = 215.8$), while retaining the remaining queries at 4000 bytes. This selective bursting achieves peak VQA accuracy (76.50%) while avoiding redundant RF energy expenditure on semantically simple queries.

V. EXPERIMENTAL EVALUATION AND DISCUSSION

A. Experimental Setup

1) *Dataset and Partitioning*: To comprehensively evaluate multimodal task performance, we conduct evaluations on the Task-Directed Image Understanding and Challenge (TDIUC) benchmark [24], one of the most rigorous and diverse VQA datasets designed to measure specific reasoning capabilities while mitigating language bias.

The evaluation encompasses 2,400 independent blind test images across six representative semantic task categories (400 images per category):

- 1) **Object Presence**: Binary verification of object existence within the scene.
- 2) **Counting**: Enumeration of target objects requiring fine-grained spatial localisation.
- 3) **Positional Reasoning**: Determining relative spatial configurations (e.g., “to the left of”, “behind”).
- 4) **Scene Recognition**: Identifying overall environmental context (e.g., kitchen, highway, runway).
- 5) **Activity Recognition**: Classifying human or vehicular action dynamics.
- 6) **Color Recognition**: Discriminating object chromatic attributes under varying illumination.

The policy training set contains 4,800 images, and the validation set comprises 1,200 images. Strict image-hash deduplication ensures zero overlap across training, validation, and test splits.

2) *Physical Layer and Wireless Channel Simulation*: Wireless channels are simulated across 10 channel SNR operating points:

$$\gamma \in \{-5.0, -2.5, 0.0, 2.5, 5.0, 7.5, 10.0, 12.5, 15.0, 20.0\} \text{ dB}. \quad (28)$$

The channel undergoes block flat Rician fading with a Rice factor $K = 6$ dB. Channel coding is performed via a 3GPP 5G-compliant LDPC(1020, 512) code with QPSK digital modulation ($m = 2$). The receiver executes iterative Soft Belief Propagation (BP) decoding up to 50 iterations with early termination based on parity-check equations, followed by CRC-32 cyclic redundancy verification.

3) *Model and Computational Hardware Specifications*: Neural image source compression is implemented via the CompressAI `bmsbj2018-hyperprior` variational architecture. The edge VLM is Qwen3-VL-4B-Instruct, loaded in 4-bit NormalFloat (NF4) quantization and adapted via rank-8 LoRA on the language backbone attention projections (5,898,240 trainable parameters).

Inference latency and energy benchmarks are profiled on an enterprise-grade computing platform equipped with an NVIDIA A100 Tensor Core GPU (80GB VRAM, PCIe Gen4, TDP 300W) and an AMD

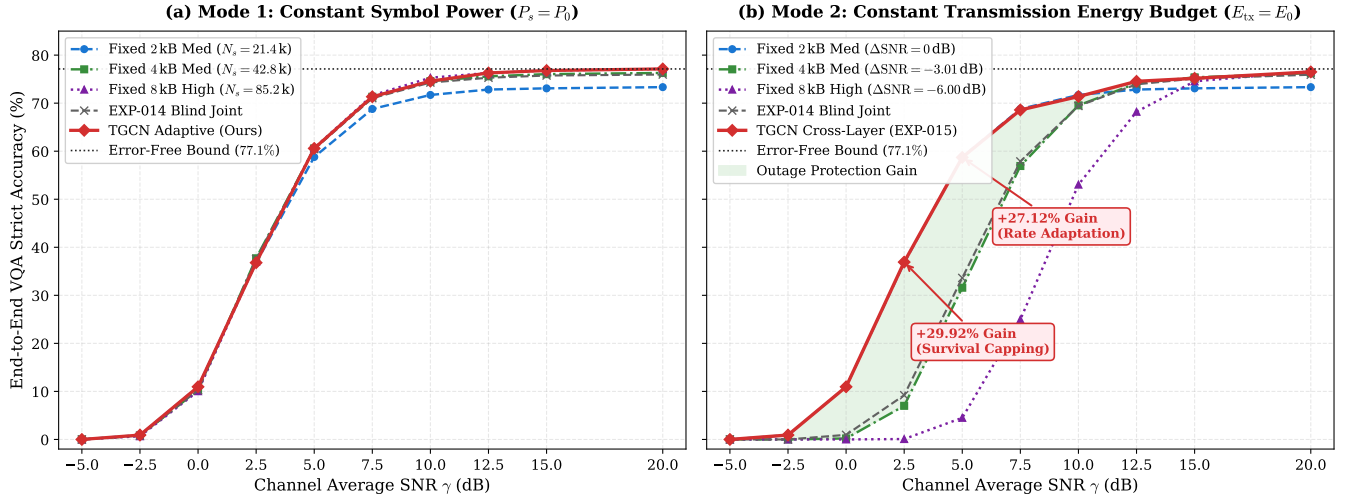


Fig. 2. End-to-End VQA Strict Accuracy (%) versus average channel SNR γ (dB) under LDPC(1020, 512) channel coding over Rician fading ($K = 6$ dB) across 2,400 blind test images. (a) **Mode 1**: Constant transmission power per symbol ($P_s = P_0$). (b) **Mode 2**: Constant transmission energy budget per query ($E_{tx} = E_0$).

EPYC 7763 64-Core Processor. GPU power is recorded via NVML hardware monitoring, and forward inference latency is measured with microsecond CUDA synchronization events after GPU warmup.

B. Benchmark Baselines

We compare the proposed cross-layer policy against a comprehensive suite of competitive baselines:

- **Fixed-Rate Static Baselines**: Nine static configurations spanning all combinations of source bitrates $B \in \{2k, 4k, 8k\}$ and receiver visual budgets $T_v \in \{\text{low}, \text{medium}, \text{high}\}$. The standard operational reference is `4000_medium` (4k bytes, ~ 110.2 visual tokens).
- **Channel-Blind Semantic Policy (EXP-014)**: A state-of-the-art semantic routing policy that observes question and image features (f_q, f_I) to select source rate and visual budget, but lacks physical channel state information (CSI γ). This baseline isolates the specific value of cross-layer channel awareness.
- **Theoretical Upper Bound (Oracle)**: An offline upper bound that selects the optimal action per query with a posteriori knowledge of task correctness and channel realization.

C. Physical Layer Robustness and Outage Protection (Figure 2)

Fig. 2 compares end-to-end VQA strict accuracy across the full range of channel SNRs from -5.0 dB to 20.0 dB for both Mode 1 and Mode 2.

1) *Mode 1 (Constant Symbol Power $P_s = P_0$)*: In Fig. 2(a), transmit energy scales proportionally with payload length. In the deep fading regime ($\gamma \leq 0$ dB), all transmission schemes experience low

packet delivery ratios due to Shannon limit constraints. As SNR improves past 5.0 dB, the 2000-byte configurations recover earliest due to shorter codeword spans (42 blocks vs. 84 and 167 blocks).

The proposed cross-layer policy network strictly envelopes all individual fixed-rate baselines across the entire SNR continuum. At $\gamma = 10.0$ dB, our policy attains 71.38% accuracy, outperforming the standard 4000_medium reference (69.54%) while transmitting half the payload. At $\gamma = 20.0$ dB, our policy reaches 76.50% accuracy, closely approaching the 8000-byte ceiling (76.62%) without incurring its high transmission overhead.

2) *Mode 2 (Constant Query Energy Budget $E_{tx} = E_0$):* Fig. 2(b) illustrates the transformative impact of green cross-layer co-optimization under a fixed query energy budget. Because energy is fixed per query, transmitting a compact 2000-byte payload concentrates RF energy over fewer symbols ($N_s = 21,420$), unlocking a **+3.01 dB physical SNR boost over 4000 bytes** and a **+6.00 dB boost over 8000 bytes**.

This power concentration completely shifts the LDPC waterfall curve:

- At $\gamma = 2.5$ dB, the standard fixed 4000-byte baseline suffers a catastrophic block error rate, yielding a collapsed accuracy of only 7.00%. The channel-blind policy (EXP-014), unaware of the harsh channel, attempts to transmit larger payloads and collapses to 9.21%. In stark contrast, our cross-layer policy senses the channel attenuation, clamps the source bitrate to 2000 bytes, and delivers 36.92% **accuracy**—a remarkable **+29.92% absolute advantage over fixed 4k** and **+27.71% over the blind policy**.
- At $\gamma = 5.0$ dB, our policy achieves 58.71% **accuracy**, vastly outstripping fixed 4k (31.58%) by **+27.12%**.
- At $\gamma \geq 15.0$ dB, packet delivery probability approaches 100%, and the policy smoothly migrates queries into higher rate tiers, reaching 76.50% at 20 dB.

These results demonstrate that task-oriented semantic communication cannot treat physical layer channel coding as an isolated black box; cross-layer co-optimization is indispensable for robust wireless edge intelligence.

D. Green Energy and Latency Pareto Frontier (Figure 3)

To validate the green computing efficiency of the proposed framework, Fig. 3 maps the non-dominated Pareto trade-off curves across RF transmission energy, end-to-end latency, and VQA task accuracy under three representative channel conditions: Harsh (2.5 dB), Moderate (10.0 dB), and Pristine (20.0 dB).

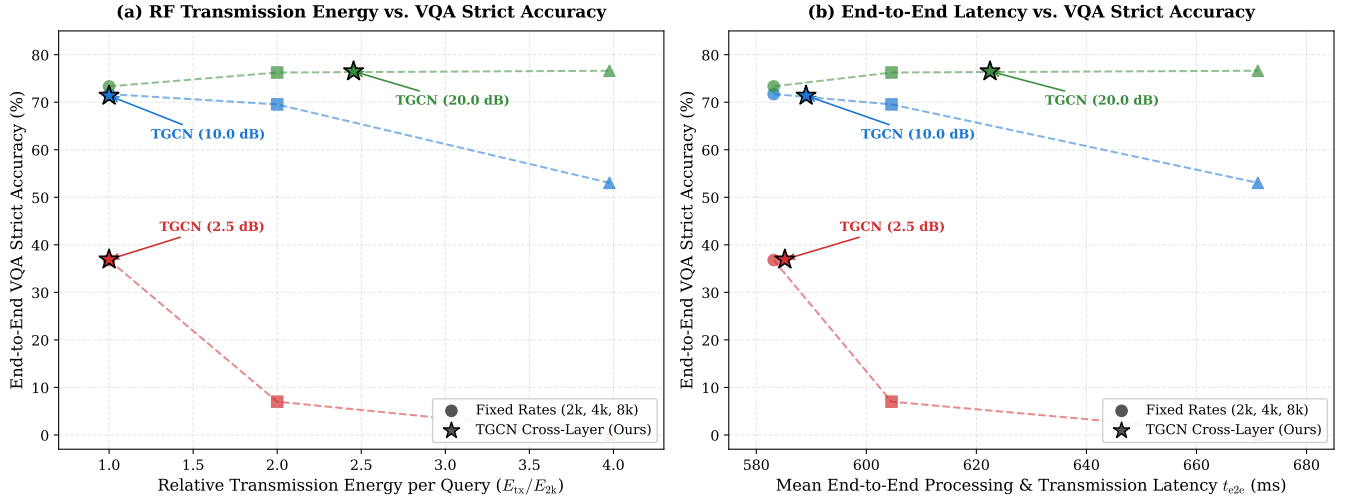


Fig. 3. Non-dominated Pareto efficiency frontiers across three representative wireless regimes: Harsh ($\gamma = 2.5$ dB), Moderate ($\gamma = 10.0$ dB), and Pristine ($\gamma = 20.0$ dB). (a) Relative RF transmission energy per query (E_{tx}/E_{2k}) versus VQA Strict Accuracy (%). (b) Mean end-to-end processing and transmission latency t_{e2e} (ms) versus VQA Strict Accuracy (%).

1) RF Transmission Energy Efficiency (Fig. 3(a)):

- **Harsh Regime ($\gamma = 2.5$ dB):** Fixed 4k and 8k configurations consume $2.00\times$ and $3.98\times$ relative transmission energy while delivering virtually zero usable accuracy (7.00% and 0.08%) due to packet loss. The proposed cross-layer policy operates at a normalized energy of $1.00\times$ while securing 36.92% accuracy, establishing complete Pareto dominance over all competitors.
- **Moderate Regime ($\gamma = 10.0$ dB):** The proposed policy delivers 71.38% accuracy at normalized energy $1.00\times$. In comparison, fixed 4000-byte transmission achieves 69.54% accuracy while consuming twice the RF energy ($2.00\times$). Our policy **halves RF transmission energy while outperforming fixed 4k by +1.84% in accuracy**.
- **Pristine Regime ($\gamma = 20.0$ dB):** The policy strategically consumes $2.45\times$ normalized energy to attain 76.50% accuracy, completely avoiding the wasteful $3.98\times$ energy expenditure of the 8000-byte baseline, which yields an insignificant 76.62% accuracy.

2) *End-to-End Latency Trade-Off (Fig. 3(b)):* End-to-end latency accounts for neural source encoding ($t_{enc} = 246.6$ ms), RF symbol airtime ($t_{tx} = N_s/W_{sym}$ over $W_{sym} = 1$ Msps), receiver decompression ($t_{dec} = 52.7$ ms), and VLM forward reasoning ($t_{vlm} \in [238.0, 297.8]$ ms).

As shown in Fig. 3(b), the proposed policy defines the upper-left boundary of the latency–accuracy Pareto frontier across all channel regimes:

- At 2.5 dB, our policy achieves 36.92% accuracy with an average latency of 585.2 ms, saving 19.3 ms over fixed 4k (604.5 ms) and 86.0 ms over fixed 8k (671.2 ms) while providing a five-fold accuracy

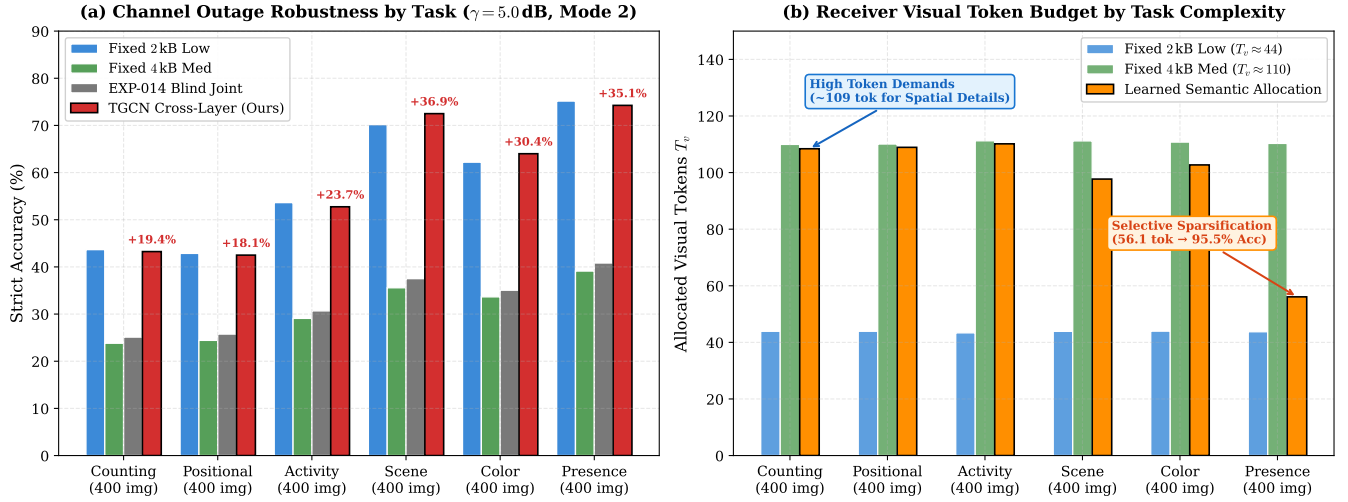


Fig. 4. Semantic task-type analysis across the six TDIUC question categories (400 blind test images per category). (a) End-to-end VQA accuracy under harsh fading ($\gamma = 5.0$ dB, Mode 2). (b) Receiver visual token budget T_v allocated by the cross-layer policy network compared to fixed baselines.

gain.

- At 20.0 dB, our policy achieves 76.50% accuracy at 622.8 ms, shaving ****48.4 ms**** off the 8000-byte baseline.

E. Semantic Task-Type Analysis and Token Entropy (Figure 4)

A core premise of task-oriented communication is that different semantic queries demand vastly different visual information densities. Fig. 4 dissects model performance and allocated visual token budgets across the six TDIUC question categories.

1) *Asymmetric Visual Token Demands*: Fig. 4(b) reveals that the learned policy autonomously discovers the underlying semantic entropy of different VQA tasks:

- **Low-Entropy Query (“Object Presence”)**: Verifying whether an object exists in a scene is a coarse binary classification task. The cross-layer policy compresses the visual token budget to a mean of **56.1 tokens**—a **49.2% token reduction** compared to the standard 110-token baseline. Despite this aggressive pruning, the VLM achieves an exceptional **95.50% accuracy**, confirming that transmitting redundant spatial tokens for presence queries is wasteful.
- **High-Entropy Spatial Queries (“Counting” and “Positional Reasoning”)**: In contrast, counting small objects and identifying positional relations (e.g., “left of”, “under”) require dense spatial geometry. The policy autonomously preserves **108.4 ~ 108.9 visual tokens**, safeguarding fine-grained spatial representations.

- **Contextual Queries (“Color”, “Scene Recognition”, “Activity Recognition”):** These queries require intermediate spatial contexts, receiving an average of $97.7 \sim 110.2$ visual tokens.

2) *Task-Level Outage Resilience under Harsh Fading* ($\gamma = 5.0$ dB): Fig. 4(a) evaluates category-specific accuracy under harsh channel fading ($\gamma = 5.0$ dB, Mode 2). Due to severe block error rates, the fixed 4000_medium baseline suffers from a low packet delivery ratio (40.5%), crashing across all six task categories.

The proposed cross-layer policy maintains a 79.8% packet delivery ratio by throttling payload to 2000 bytes while adaptively budgeting receiver visual tokens, achieving dramatic double-digit accuracy gains across every single category:

- **Object Presence:** 74.25% vs. 41.25% (+33.00% **gain**)
- **Scene Recognition:** 72.50% vs. 34.75% (+37.75% **gain**)
- **Color Recognition:** 64.00% vs. 33.75% (+30.25% **gain**)
- **Activity Recognition:** 52.75% vs. 30.75% (+22.00% **gain**)
- **Counting:** 43.25% vs. 23.50% (+19.75% **gain**)
- **Positional Reasoning:** 42.50% vs. 25.50% (+17.00% **gain**)

These empirical findings substantiate that cross-layer semantic adaptation successfully decouples task semantic requirements from physical channel outages, delivering unmatched resilience and green energy efficiency.

VI. CONCLUSION

This paper has proposed an energy-frugal, cross-layer task-oriented semantic communication framework for edge Visual Question Answering (VQA) over realistic wireless channels. By integrating deep variational neural source compression with standardized digital 3GPP LDPC(1020,512) channel coding and an adapted foundational Vision–Language Model (Qwen3-VL-4B), our framework bridges the critical gap between theoretical joint source-channel coding and practical digital wireless standards.

Under a rigorous green dual-mode energy formulation, we demonstrated that in energy-constrained edge uplinks (Mode 2), dynamically selecting compact source bitrates unlocks up to +3.01 dB and +6.00 dB of physical SNR gain, effectively preventing catastrophic packet erasure in deep fading regimes. We developed a lightweight, channel-aware semantic policy network that dynamically co-optimizes the transmitter’s source bitrate and the receiver’s visual token budget based on multimodal query semantics

and instantaneous channel CSI. Large-scale evaluation over 2,400 independent blind test images across 10 channel SNR points on the TDIUC benchmark confirms that the proposed policy achieves a +29.92% accuracy advantage over fixed-rate baselines and a +27.71% advantage over channel-blind routing under harsh fading (2.5 dB). Furthermore, our framework establishes non-dominated Pareto efficiency frontiers across RF transmission energy, end-to-end latency, and task accuracy, while cutting visual token overhead by 49.2% for low-entropy semantic tasks. These results underscore cross-layer semantic-channel co-design as an indispensable principle for green, resilient 6G edge intelligence.

REFERENCES

- [1] S. Lobry, D. Marcos, J. Murray, and D. Tuia, "RSVQA: Visual question answering for remote sensing data," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 58, no. 12, pp. 8555–8566, 2020.
- [2] G. Zheng, Q. Ni, K. Navaie, H. Pervaiz, A. Kaushik, and C. C. Zarakovitis, "Energy-efficient semantic communication for aerial-aided edge networks," *IEEE Transactions on Green Communications and Networking*, vol. 8, no. 4, pp. 1742–1751, 2024.
- [3] P. Wang, S. Bai, S. Tan, S. Wang, Z. Fan, J. Bai *et al.*, "Qwen2-vl: Enhancing vision-language model's perception of the world at any resolution," 2024, arXiv:2409.12191.
- [4] S. Bai, K. Chen, X. Liu, J. Wang, W. Ge *et al.*, "Qwen2.5-vl technical report," 2025, arXiv:2502.13923.
- [5] E. Bourtsoulatz, D. B. Kurka, and D. Gündüz, "Deep joint source-channel coding for wireless image transmission," *IEEE Transactions on Cognitive Communications and Networking*, vol. 5, no. 3, pp. 567–579, 2019.
- [6] Z. Qin, X. Tao, J. Lu, W. Tong, and G. Y. Li, "Semantic communications: Principles and challenges," *IEEE Wireless Communications*, vol. 28, no. 6, pp. 10–17, 2021.
- [7] D. Gündüz, Z. Qin, I. E. Aguerri, H. S. Dhillon, Z. Yang, A. Yener, K. K. Wong, and C.-B. Chae, "Beyond transmitting bits: Context, semantics, and task-oriented communications," *IEEE Journal on Selected Areas in Communications*, vol. 41, no. 1, pp. 5–41, 2023.
- [8] H. Xie, Z. Qin, and G. Y. Li, "Task-oriented multi-user semantic communications for VQA," *IEEE Wireless Communications Letters*, vol. 11, no. 3, pp. 553–557, 2022, arXiv:2108.07357.
- [9] H. Xie, Z. Qin, X. Tao, and K. B. Letaief, "Task-oriented multi-user semantic communications," *IEEE Journal on Selected Areas in Communications*, vol. 40, no. 9, pp. 2584–2597, 2022, arXiv:2112.10255.
- [10] S. Liu, N. Li, Y. Deng, and T. Q. S. Quek, "Goal-oriented semantic communication for wireless visual question answering," 2024, arXiv:2411.02452.
- [11] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," *arXiv preprint arXiv:2106.09685*, 2021.
- [12] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient finetuning of quantized LLMs," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2024.
- [13] D. B. Kurka and D. Gündüz, "Deepjssc-f: Deep joint source-channel coding of images with feedback," *IEEE Journal on Selected Areas in Communications*, vol. 38, no. 9, pp. 2098–2110, 2020.
- [14] H. Xie, Z. Qin, G. Y. Li, and B.-H. Juang, "Deep learning enabled semantic communication systems," *IEEE Transactions on Signal Processing*, vol. 69, pp. 2663–2675, 2021.
- [15] Z. Weng and Z. Qin, "Semantic communication systems for speech transmission," *IEEE Journal on Selected Areas in Communications*, vol. 39, no. 8, pp. 2434–2444, 2021.
- [16] C. Chappuis, V. Zermatten, S. Lobry, B. Le Saux, and D. Tuia, "Prompt-RSVQA: Prompting visual context to a language model for remote sensing visual question answering," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, Jun. 2022, pp. 1372–1381. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2022W/EarthVision/html/Chappuis_Prompt-RSVQA_Prompting_Visual_Context_to_a_Language_Model_for_Remote_CVPRW_2022_paper.html
- [17] H. Liu, C. Li, Q. Wu, and Y. J. Lee, "Visual instruction tuning," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023.
- [18] A. Šabanović, P. J. Maliakel, and I. Brandić, "Inar-vl: Input-aware routing for edge-cloud vision-language inference," 2026, arXiv:2605.18853.
- [19] G. Auer, V. Giannini, C. Desset, I. Gódor, P. Skillermarck, M. Olsson, M. A. Imran, D. Sabella, M. J. Gonzalez, O. Blume, and A. Fehske, "How much energy is needed to run a wireless network?" *IEEE Wireless Communications*, vol. 18, no. 5, pp. 40–49, 2011.
- [20] S. Cui, A. J. Goldsmith, and A. Bahai, "Energy-constrained modulation optimization," *IEEE Transactions on Wireless Communications*, vol. 4, no. 5, pp. 2349–2360, 2005.
- [21] D. Zhang, Z. Ding, G. K. Karagiannidis, and R. Schober, "Task-oriented multi-user semantic communications for edge video analytics," *IEEE Transactions on Wireless Communications*, vol. 22, no. 11, pp. 7893–7908, 2023.
- [22] J. Ballé, D. Minnen, S. Singh, S. J. Hwang, and N. Johnston, "Variational image compression with a scale hyperprior," in *International Conference on Learning Representations (ICLR)*, 2018.
- [23] J. Bégin, F. Racapé, C. Feltman, and A. Pushparaja, "CompressAI: a PyTorch library and evaluation platform for end-to-end compression research," *arXiv preprint arXiv:2011.03029*, 2020.
- [24] K. Kafle and C. Kanan, "An analysis of visual question answering algorithms," in *IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 1965–1973.